Auto-Clip CLAHE: Noise-Aware Automatic Clip-Limit Selection for Adaptive Contrast Enhancement

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***Abstract*—Contrast Limited Adaptive Histogram Equalization (CLAHE) is a workhorse of local contrast enhancement, but its single most important parameter—the clip limit—is normally fixed by hand. A clip limit that produces vivid, detailed results on a clean image will amplify sensor noise into visible artifacts when the same setting is applied to a noisy image, while a clip limit chosen to be safe under noise leaves clean images under-enhanced. We propose *Auto-Clip CLAHE*, a parameter-light method that selects a *per-tile* clip limit automatically from two image statistics: the global noise floor, estimated robustly from the diagonal wavelet sub-band using the median-absolute-deviation (MAD) estimator, and the local Shannon entropy of each tile. The clip limit grows with local texture and shrinks as the estimated noise rises, so the method enhances aggressively on clean, detail-rich regions and backs off automatically as noise increases. On the 24-image Kodak suite, degraded with reduced contrast and additive Gaussian noise at six levels ($\sigma \in \{0, \ldots, 25\}$), Auto-Clip behaves like an aggressive fixed CLAHE on clean inputs (entropy gain 1.16 vs. 1.18 bits) yet attains the *lowest* no-reference BRISQUE score among all enhancement methods at the highest noise level (71.0 vs. 76.4 and 85.8 for the fixed baselines), together with the best structural stability (SSIM) under strong noise. A two-parameter grid search calibrated on a six-image subset selects the operating point with a single parameter-free objective, and an ablation confirms that the noise-normalisation term is the source of the robustness: removing it collapses the method to the over-enhancing fixed-high baseline. The complete implementation and reproduction scripts are publicly available.**

***Index Terms*—contrast enhancement, CLAHE, adaptive histogram equalization, clip limit, noise estimation, wavelet MAD, image quality assessment, BRISQUE.**

I. Introduction

Histogram-based contrast enhancement is among the oldest and most widely used families of image-processing techniques. Global Histogram Equalization (GHE) remaps intensities so that the output histogram is approximately uniform, but because it uses a single global mapping it cannot adapt to spatially varying content and often over-stretches some regions while leaving others flat. Adaptive Histogram Equalization (AHE) addresses this by computing a separate mapping for each local tile [1], dramatically improving local contrast—at the cost of strongly amplifying noise in near-homogeneous regions. Contrast Limited Adaptive Histogram Equalization (CLAHE) [2] tames this behaviour by *clipping* each tile histogram at a fixed ceiling and redistributing the clipped mass, which bounds the local slope of the transfer function and therefore the maximum contrast (and noise) amplification. CLAHE remains a default choice in medical imaging, photography, and computer-vision pre-processing, and efficient real-time realizations are well established [3].

The Achilles' heel of CLAHE is precisely the parameter that gives it its power: the clip limit. The clip limit is a single scalar that must trade off enhancement against artifact amplification, and the *best* value depends on the image. A low clip limit is conservative—safe on noisy inputs but timid on clean ones—whereas a high clip limit produces striking detail on clean images but turns noise into blotches and halos once the input is degraded. In practice the clip limit is tuned manually for a fixed acquisition setting, which breaks down whenever the noise level is unknown or varies between images, as it does across cameras, ISO settings, and lighting conditions.

This paper asks a simple question: *can the clip limit be chosen automatically, per image and per region, from quantities the algorithm can measure directly?* We answer in the affirmative with *Auto-Clip CLAHE*. The method couples two ingredients. First, it estimates the global noise floor σ using the classical wavelet MAD estimator of Donoho and Johnstone [4], which reads the noise level off the finest-scale diagonal detail coefficients and is robust to image structure. Second, it measures the local Shannon entropy $E_{i,j}$ of every tile as a texture-energy descriptor. The per-tile clip limit is then

$$C_{i,j} = \alpha + \beta \, \frac{E_{i,j}}{\sigma + \varepsilon}, \tag{1}$$

so that detail-rich tiles receive a higher clip limit (more enhancement) while a rising noise estimate suppresses the clip limit everywhere (more caution). The two coefficients (α, β) are global and fixed once by a small calibration search; the spatial and noise adaptation is entirely automatic.

Our contributions are:

- A noise-aware, per-tile automatic clip-limit rule (Eq. 1) that requires no manual re-tuning when the input noise level changes.
- A custom CLAHE engine that accepts a distinct clip limit for every tile with bilinear interpolation between tile mappings, since standard libraries expose only a single global clip limit.

- A parameter-free calibration objective $J = G - \lambda P$ that balances mean entropy gain against the mean no-reference quality penalty, with λ set automatically from the dispersion of the two terms.
- A reproducible evaluation on the Kodak suite across six noise levels against GHE and two fixed-CLAHE baselines, plus an ablation isolating the contribution of the noise term.

The remainder of the paper reviews related work (Section II), details the method and experimental protocol (Section III), reports results (Section IV), and concludes (Section V).

II. Related Work

Histogram equalization and its adaptive variants. GHE is attractive for its simplicity but is well known to over-enhance and to ignore local structure. AHE [1] introduced tile-wise mappings and identified, in the same work, the central drawback that motivates our study: AHE over-amplifies noise in homogeneous regions. The contrast-limited variant CLAHE [2] bounds amplification by clipping tile histograms before computing their cumulative distribution and interpolating between neighbouring tile mappings to avoid block boundaries. These ideas are now standard, and real-time implementations target embedded and clinical use [3]. All of these methods, however, treat the clip limit as a user-supplied constant.

Adaptive parameter selection. A number of works adjust enhancement strength using local statistics such as variance or gradient activity, typically to push enhancement harder in textured regions. What is comparatively rare is to make the strength explicitly a function of the *estimated noise level*, so that the same algorithm becomes more conservative as the input degrades. Our rule does exactly this by placing a noise estimate in the denominator of the clip-limit expression.

Noise estimation. Estimating the noise level of a single image is itself a classical problem. Fast Laplacian-based estimators [5] are simple but can be biased by image structure. The wavelet MAD estimator of Donoho and Johnstone [4], originally developed for wavelet shrinkage denoising, reads the noise standard deviation from the median absolute deviation of the finest-scale diagonal coefficients and is robust because high-frequency detail occupies a small fraction of that subband. We adopt it as a lightweight, structure-robust noise floor.

Image quality assessment. Because enhancement has no single "ground-truth" output, we evaluate with complementary measures. The Structural Similarity Index (SSIM) [6] captures structural agreement with a reference, while BRISQUE [7] is a no-reference model that scores the "naturalness" of an image from spatial natural-scene statistics, with lower scores indicating fewer distortions. BRISQUE is particularly suitable here because it penalises the blotchy, unnatural artifacts that over-aggressive enhancement produces on noisy inputs, exactly the failure mode we wish to avoid. Standardised distorted-image databases such as TID2013 [8] underpin the design and validation of such metrics.

III. Materials and Methods

A. Dataset and Degradation Model

We use the 24-image Kodak Lossless True Color Image Suite [9], a standard benchmark of natural photographs (512×768). To create a controlled enhancement problem we degrade each image in two stages. First, contrast is reduced by compressing the lightness channel toward mid-grey,

$$L' = 128 + (L - 128)\,\kappa, \qquad \kappa = 0.7, \tag{2}$$

simulating a low-contrast acquisition. Second, zero-mean additive Gaussian noise of standard deviation $\sigma_n \in \{0, 5, 10, 15, 20, 25\}$ (on the 0–255 scale) is added to all channels. For every (image, noise-level) pair a single noisy realisation is generated with a fixed seed and presented to *all* methods, so that every comparison is paired.

B. Global Noise-Floor Estimation

The noise level is estimated on the lightness channel using the wavelet MAD estimator [4]. A single-level Haar (`db1`) 2-D discrete wavelet transform is applied and the diagonal (HH) detail coefficients w_d are extracted; the noise standard deviation is

$$\hat{\sigma} = \frac{\operatorname{median}\big(\,|w_d - \operatorname{median}(w_d)|\,\big)}{0.6745}. \tag{3}$$

The constant 0.6745 rescales the MAD to a standard deviation for Gaussian data. Because edges and texture contribute only sparsely to the HH band, the robust spread of w_d is dominated by noise, yielding a stable global floor.

C. Local Texture Energy

Each tile's texture energy is its Shannon entropy. For a tile with normalised 256-bin intensity histogram p,

$$E_{i,j} = -\sum_{k=0}^{255} p_k \log_2 p_k, \tag{4}$$

which is large for detailed, high-information tiles and small for flat ones. Entropy is used because it responds to the *distribution* of intensities rather than their absolute spread; a variance-based descriptor is evaluated as an alternative in the ablation.

D. Adaptive Clip-Limit Rule

The proposed per-tile clip limit combines the two statistics:

$$C_{i,j} = \operatorname{clip}\left(\alpha + \beta\,\frac{E_{i,j}}{\hat{\sigma} + \varepsilon},\; C_{\min},\, C_{\max}\right), \tag{5}$$

with $\varepsilon = 1$ guarding against division by zero and the result clamped to $[C_{\min}, C_{\max}] = [0.001, 0.05]$ for numerical stability. Here $C_{i,j}$ is a *normalised* clip limit: a fraction of the tile's pixel count at which each histogram bin is capped. The structure of Eq. 5 encodes the design intent directly. Holding noise fixed, tiles with more texture receive a larger clip limit and hence stronger enhancement. Holding content fixed, a larger estimated noise floor shrinks the clip limit *globally*, so the method automatically becomes more conservative on noisy inputs. The offset α sets a baseline clip limit and the gain β controls how strongly the texture-to-noise ratio modulates it.

E. Per-Tile CLAHE Engine

Standard CLAHE implementations expose only one global clip limit, so we implement a CLAHE engine that accepts a full clip map. For each tile a histogram is built, capped at $C_{i,j} \cdot$ (tile pixels), and the clipped excess is redistributed uniformly across all bins before the cumulative distribution function is formed into a 256-entry look-up table [2]. The four tile mappings nearest to each pixel are then combined by bilinear interpolation, which removes the block boundaries that naive tiling would introduce. All enhancement is performed on the L channel of the CIELAB representation and the chrominance channels are left untouched, so colours are preserved and only lightness contrast is modified.

F. Baselines

We compare against three baselines that share the same enhancement substrate wherever possible, to isolate the effect of the clip-limit policy rather than implementation differences:

- **GHE** – global histogram equalization of the L channel.
- **Fixed-CLAHE-low** – our engine with a constant clip limit $C = 0.01$ on every tile (a conservative setting).
- **Fixed-CLAHE-high** – our engine with a constant clip limit $C = 0.04$ (an aggressive setting).

The two fixed-CLAHE baselines and Auto-Clip use the *identical* engine and 8×8 tile grid; only the clip map differs, making the comparison a direct test of adaptive versus constant clipping.

G. Evaluation Metrics

Enhancement quality is assessed with three complementary measures:

- **Entropy gain** ΔH – the increase in L-channel Shannon entropy from input to output; a direct, reference-free measure of how much information/contrast the method adds. Higher is more enhancement.
- **BRISQUE** [7] – a no-reference quality score where *lower* is better; it penalises the unnatural statistics produced by amplified noise, and therefore measures artifact severity.
- **SSIM** [6] – computed between a method's output on the *noisy* input and the same method's output on the *clean* (noise-free, low-contrast) input. This measures how stable a method's enhancement is under noise: a method that yields the same result with and without noise scores high, while one that lets noise reshape its output scores low. PSNR against the same reference is also reported.

This combination is deliberate: ΔH rewards enhancement, BRISQUE punishes artifacts, and SSIM measures robustness. A method that simply does nothing scores low on ΔH; a method that over-enhances noise scores poorly on BRISQUE and SSIM. A good method must do well on all three.

H. Implementation and Reproducibility

The method is implemented in Python using OpenCV [10], NumPy [11], scikit-image [12], and PyWavelets, with BRISQUE provided by the OpenCV `quality` module using the LIVE-trained model. All randomness is seeded; every figure and table in this paper is regenerated by the released scripts [13].

TABLE I
CALIBRATION OBJECTIVE $J = G - \lambda P$ OVER THE (α, β) GRID (SIX CALIBRATION IMAGES, ALL NOISE LEVELS). HIGHER IS BETTER; THE MAXIMUM IS SHOWN IN BOLD.

	β				
α	0.005	0.010	0.020	0.030	0.040
0.000	0.198	0.286	**0.325**	0.291	0.250
0.002	0.261	0.316	0.315	0.270	0.234
0.005	0.312	0.321	0.284	0.242	0.218

IV. EXPERIMENTAL RESULTS

A. Hyperparameter Calibration

The two global coefficients (α, β) are selected by a grid search on a fixed six-image calibration subset (`kodim01, 05, 07, 13, 19, 23`) disjoint in spirit from the held-out analysis, sweeping $\alpha \in \{0, 0.002, 0.005\}$ and $\beta \in \{0.005, 0.01, 0.02, 0.03, 0.04\}$. For each setting we measure, averaged over the calibration images and all six noise levels, the mean entropy gain G and a mean no-reference quality penalty P (the rise in BRISQUE relative to the noisy input). Because raw enhancement and raw artifact penalty live on very different scales, we combine them with a single parameter-free objective

$$J = G - \lambda P, \qquad \lambda = \frac{\mathrm{std}(G)}{\mathrm{std}(P)}, \tag{6}$$

where λ is fixed automatically from the dispersion of the two terms across the grid (here $\lambda = 0.0321$), so no weighting is hand-chosen. This objective rewards enhancement while subtracting the artifact cost it incurs.

Table I and Fig. 1 show a single, clean maximum at $(\alpha, \beta) = (0.0,\ 0.02)$ with $J = 0.325$. At this operating point the mean entropy gain is $G = 0.78$ bits against an artifact penalty of $P = 14.1$ BRISQUE points. Pushing β higher keeps adding enhancement (G rises to 0.89 at $\beta = 0.04$) but the artifact penalty rises faster ($P = 19.9$), so J falls—the objective correctly identifies the point of diminishing returns rather than the most aggressive setting. We use $(\alpha, \beta) = (0.0, 0.02)$ for all subsequent experiments.

B. Main Comparison

Tables II–IV report results averaged over all 24 Kodak images at the calibrated operating point, and Fig. 2 summarises the three metrics as a function of noise level.

The tables tell a consistent story. On *clean* images ($\sigma_n = 0$), Auto-Clip behaves like an aggressive enhancer: its entropy gain of 1.16 bits is essentially tied with Fixed-CLAHE-high (1.18) and far above Fixed-CLAHE-low (0.81) and GHE (0.11). At this point its BRISQUE (22.8) is close to the aggressive baseline (24.2), which is the expected cost of strong enhancement on an already-clean image.

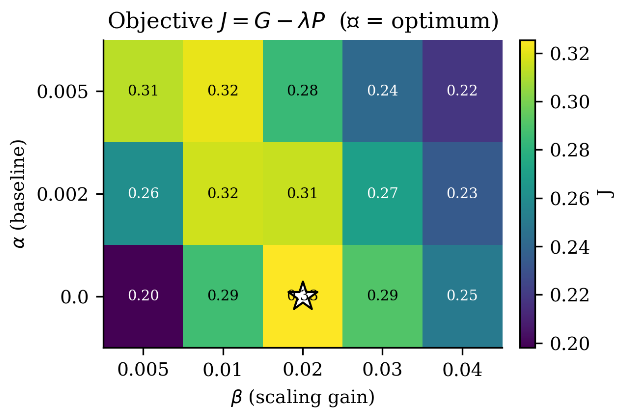

Fig. 1. Calibration objective J over the (α, β) grid. The optimum ($\star$) sits at $(\alpha, \beta) = (0.0, 0.02)$; the surface is smooth and single-peaked.

TABLE II
BRISQUE (LOWER IS BETTER), MEAN OVER 24 IMAGES, BY NOISE LEVEL σ_n. BEST (LOWEST) *enhancement* METHOD AT EACH LEVEL IN BOLD.

Method	0	5	10	15	20	25
GHE	16.1	39.2	53.0	61.1	66.7	71.4
Fixed-CLAHE-low	**14.6**	42.4	57.1	65.7	71.6	76.4
Fixed-CLAHE-high	24.2	53.8	69.4	78.0	82.5	85.8
Auto-Clip (ours)	22.8	48.3	59.5	65.2	68.6	**71.0**

TABLE III
ENTROPY GAIN ΔH IN BITS (HIGHER IS MORE ENHANCEMENT), MEAN OVER 24 IMAGES, BY NOISE LEVEL σ_n.

Method	0	5	10	15	20	25
GHE	0.11	0.11	0.09	0.07	0.06	0.04
Fixed-CLAHE-low	0.81	0.70	0.69	0.68	0.65	0.62
Fixed-CLAHE-high	1.18	1.05	0.99	0.89	0.80	0.71
Auto-Clip (ours)	1.16	0.92	0.79	0.67	0.57	0.48

As noise grows, the behaviour diverges sharply. The fixed-high baseline keeps enhancing hard ($\Delta H = 0.71$ at $\sigma_n = 25$) but pays for it with the worst BRISQUE at every noise level, reaching 85.8—it is amplifying noise into artifacts. Auto-Clip instead *reduces* its own enhancement as the noise estimate climbs (ΔH falls from 1.16 to 0.48), and the payoff is decisive: at $\sigma_n = 25$ it attains a BRISQUE of 71.0, the lowest of *any* enhancement method and below even the conservative fixed-low baseline (76.4) and the aggressive one (85.8). The SSIM results (Table IV) corroborate this: for $\sigma_n \geq 15$ Auto-Clip is the most structurally stable method overall, indicating that its output is shaped by genuine image content rather than by the noise realisation.

Crucially, *no single fixed clip limit reproduces this profile*. The low-clip setting is reasonable under noise but timid on clean images; the high-clip setting is excellent on clean images but the worst under noise. Auto-Clip occupies the favourable diagonal of both regimes by transitioning from aggressive to conservative automatically, without any change of parameters.

TABLE IV
SSIM BETWEEN EACH METHOD'S NOISY-INPUT AND CLEAN-INPUT OUTPUTS (HIGHER IS MORE ROBUST), MEAN OVER 24 IMAGES. BEST OVERALL AT EACH $\sigma_n > 0$ IN BOLD. AT $\sigma_n = 0$ THE NOISY INPUT EQUALS THE CLEAN INPUT, SO EVERY METHOD SCORES 1.000 BY CONSTRUCTION; THE COLUMN IS REPORTED FOR COMPLETENESS.

Method	0	5	10	15	20	25
GHE	1.000	**0.761**	**0.581**	0.461	0.378	0.318
Fixed-CLAHE-low	1.000	0.753	0.557	0.439	0.360	0.303
Fixed-CLAHE-high	1.000	0.692	0.528	0.432	0.366	0.316
Auto-Clip (ours)	1.000	0.728	0.565	**0.463**	**0.393**	**0.341**

TABLE V
ABLATION, MEAN OVER 24 IMAGES. BRISQUE AT $\sigma_n = 0$ AND $\sigma_n = 25$, AND ENTROPY GAIN AT $\sigma_n = 25$.

Variant	BRISQUE@0	BRISQUE@25	ΔH@25
Entropy (full)	22.8	71.2	0.48
Variance (full)	22.9	71.0	0.48
No noise term	25.2	86.0	0.71

Fig. 3 visualises the mechanism. On a clean image the clip map allocates high limits to textured tiles and low limits to flat sky; when the same image is heavily noised, the spatial pattern is retained but the entire map is pulled down, which is exactly the global caution introduced by the $\hat{\sigma}$ term in the denominator of Eq. 5. The effect of this clip map on the image itself is shown in Fig. 4: alongside the map, the noisy low-contrast input and the Auto-Clip output are placed side by side at $\sigma_n = 20$. The output recovers contrast on the building, fence, and life-ring while the suppressed clip limit keeps the flat sky from breaking up into the blotches that an aggressive fixed limit would produce. A broader qualitative comparison against the baselines is shown in Fig. 5.

C. Ablation Study

Two design choices are examined in Table V: the texture descriptor (Shannon entropy vs. log-variance) and, most importantly, the presence of the noise-normalisation term.

Replacing entropy with a log-variance texture descriptor makes essentially no difference (BRISQUE@25 of 71.0 vs. 71.2; identical ΔH), indicating that the method is insensitive to the exact texture measure and that any reasonable local activity descriptor would serve. The decisive component is the noise term. Removing it—i.e. using $C_{i,j} = \alpha + \beta E_{i,j}$ with no $\hat{\sigma}$ in the denominator—causes the method to collapse onto the over-enhancing fixed-high regime: entropy gain at $\sigma_n = 25$ jumps back to 0.71 (matching Fixed-CLAHE-high exactly) and BRISQUE@25 degrades from 71.2 to 86.0. This is direct evidence that the noise normalisation, not the spatial adaptation alone, is what delivers the robustness reported in Section IV. Fig. 6 shows the same conclusion graphically.

D. Runtime

Table VI reports the per-image processing time, measured on the 512×768 working resolution and averaged over the test

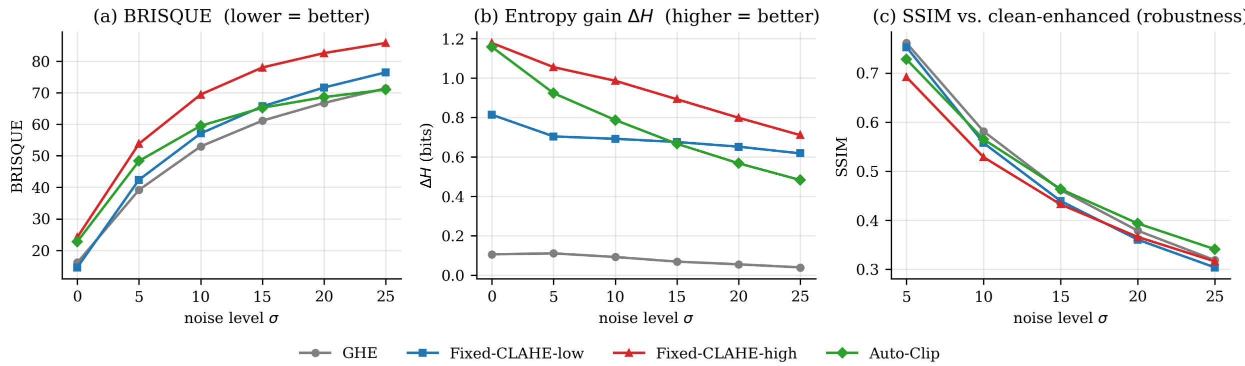

Fig. 2. Behaviour of the four methods as the input noise level grows. **Left:** BRISQUE (lower is better)—Auto-Clip starts aggressive but overtakes every other enhancer as noise rises, achieving the lowest score at $\sigma_n = 25$. **Middle:** entropy gain—Auto-Clip matches the aggressive fixed-high baseline on clean images and then deliberately backs off. **Right:** SSIM stability—Auto-Clip is the most robust enhancer at strong noise.

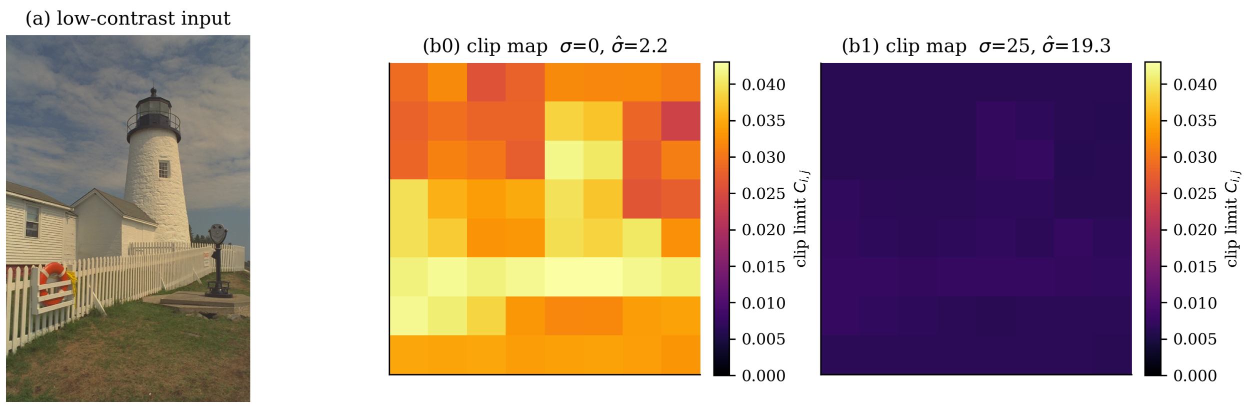

Fig. 3. Per-tile clip maps produced by Auto-Clip on `kodim19`. **Left:** input. **Centre:** clip map at $\sigma_n = 0$—high clip limits concentrate on textured regions. **Right:** clip map at $\sigma_n = 25$—the same spatial pattern is preserved but the overall level is suppressed, illustrating simultaneous spatial and noise-driven adaptation.

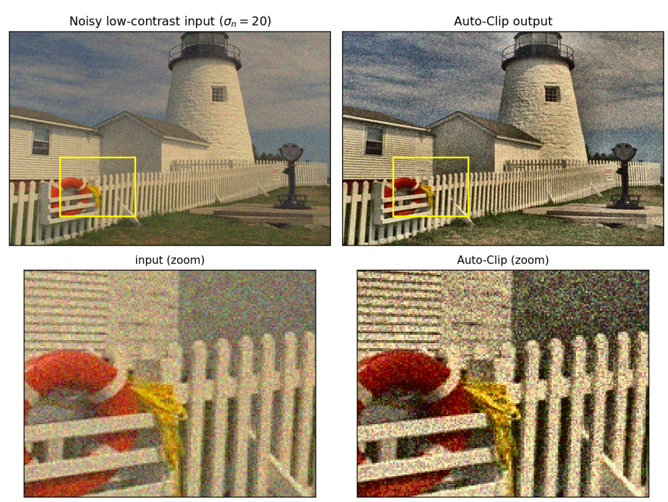

Fig. 4. Input vs. output for the clip map of Fig. 3, on `kodim19` at $\sigma_n = 20$. **Top:** full noisy low-contrast input and Auto-Clip output (crop region boxed). **Bottom:** zoom of the boxed region—contrast on the fence and life-ring is restored while noise remains controlled.

images. The reference Auto-Clip implementation runs in about 44 ms per image. Most of this cost is *not* the adaptive logic: the same custom tiling engine run with a constant clip limit (Fixed-CLAHE) already takes 34 ms, so the noise estimation, per-tile entropy, and clip-map construction add only $\approx$10 ms (23%) on top of it, of which the wavelet noise estimate is $\approx$7 ms. The optimized C++ CLAHE shipped with OpenCV completes in 6.6 ms; the gap is therefore dominated by our unvectorised, pure-Python per-tile engine rather than by the adaptivity, and a compiled or vectorised port of the engine would close most of it while leaving the method unchanged. The adaptive overhead itself is modest and runs comfortably in real time for offline and interactive use.

E. Limitations

The adaptive policy is a trade-off, not a free lunch. At the most extreme noise level the same caution that protects BRISQUE drives Auto-Clip's entropy gain *below* that of the conservative fixed-low baseline (Table III): the method chooses to under-enhance rather than risk amplifying heavy noise. Whether this is desirable depends on the application. The noise model studied here is signal-independent additive

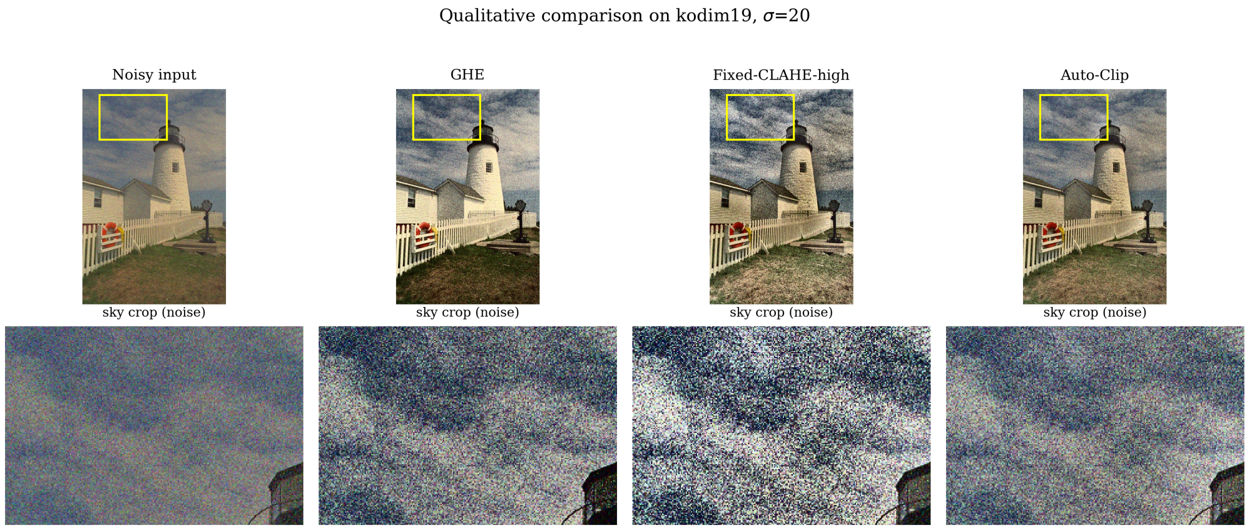

Fig. 5. Qualitative comparison on a `kodim19` crop at $\sigma_n = 20$. Fixed-CLAHE-high amplifies noise into visible grain, whereas Auto-Clip preserves contrast with markedly fewer artifacts.

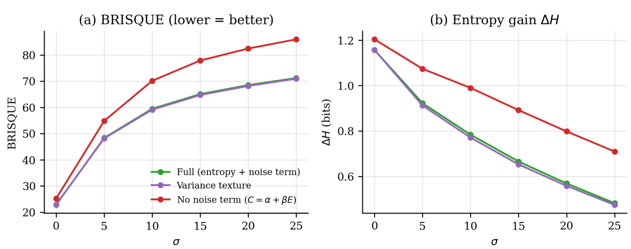

Fig. 6. Ablation across noise levels. Removing the noise-normalisation term ("no noise term") tracks the over-enhancing baseline and degrades sharply under noise, while the entropy and variance variants of the full method are nearly indistinguishable.

TABLE VI
PER-IMAGE RUNTIME AT 512×768 (MEAN OVER TEST IMAGES). THE ADAPTIVE OVERHEAD OF AUTO-CLIP OVER THE SAME ENGINE RUN WITH A FIXED CLIP IS ONLY $\approx$10 MS; THE REMAINING GAP TO OPENCV IS IMPLEMENTATION, NOT METHOD.

Method	Time (ms)
OpenCV CLAHE (standard, C++)	6.6
Fixed-CLAHE (paper engine, Python)	33.6
Auto-Clip (ours, Python)	43.7
of which: noise estimation	6.9

Gaussian noise; signal-dependent (e.g. Poisson) noise and compression artifacts are not evaluated, and the global $\hat{\sigma}$ assumes spatially uniform noise. Finally, (α, β) are calibrated on natural photographs and may need re-calibration for very different domains such as medical or astronomical imagery. In particular, the calibrated coefficients should not be assumed to transfer unchanged to real low-light captures: such images combine signal-dependent photon (Poisson) noise, non-uniform noise across the sensor, and in-camera processing, none of which are present in our additive-Gaussian Kodak protocol. The *form* of the rule—tying the clip limit inversely to an estimated noise level—should carry over, but β (and possibly the noise estimator) would likely need re-fitting on a small calibration set drawn from the target domain; validating this transfer on a real low-light dataset is left to future work.

V. Conclusion

We presented Auto-Clip CLAHE, a noise-aware method that selects the CLAHE clip limit automatically and per-tile from the local texture energy and a robust wavelet estimate of the global noise floor. On the Kodak suite across six noise levels the method enhances clean images as strongly as an aggressive fixed CLAHE, yet achieves the best no-reference quality and the best structural stability of any enhancer under strong noise—a profile that no single fixed clip limit attains. An ablation pinpoints the noise-normalisation term as the source of this robustness. The method adds essentially no tuning burden: two global coefficients are fixed once by a parameter-free calibration objective and never touched again. Future work includes signal-dependent noise models, spatially varying noise estimation, and learning the coefficients per domain. The full implementation and reproduction scripts are publicly available [13].

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